Documentation: Read a Text File from GitHub Repository into Python in Google Colab

Objective

This documentation explains how to read a text file stored in a GitHub repository directly into Python using **Google Colab**.

Introduction

Google Colab is a cloud-based Python notebook platform where you can run Python code without installing software.

Sometimes your text file is stored in a GitHub repository, and you want to read that file directly into Colab for:

- Data Analysis
- File Processing
- NLP Projects
- Machine Learning Datasets
- Practice Work

Methods to Read File from GitHub

There are **3 Common Methods**:

1. Using Raw GitHub URL (**Best Method**)
2. Using requests library
3. Cloning Full Repository using git

Method 1: Using Raw GitHub URL (Recommended)

Step 1: Open GitHub File

Example GitHub File:

https://github.com/username/repository/blob/main/data.txt

Step 2: Convert into Raw URL

Replace:

- github.com → raw.githubusercontent.com
- Remove /blob

Final URL:

https://raw.githubusercontent.com/username/repository/main/data.txt

Step 3: Read File in Google Colab

```
url = "https://raw.githubusercontent.com/username/repository/main/data.txt"

with open("/content/sample.txt", "wb") as f:
    import requests
    f.write(requests.get(url).content)

with open("/content/sample.txt", "r") as file:
    data = file.read()

print(data)
```

Short Direct Method

```
import pandas as pd

url = "https://raw.githubusercontent.com/username/repository/main/data.txt"

data = pd.read_csv(url, header=None)

print(data)
```

Method 2: Using requests Library

```
import requests

url = "https://raw.githubusercontent.com/username/repository/main/data.txt"

response = requests.get(url)

text = response.text

print(text)
```

Method 3: Clone Full GitHub Repository

Use when multiple files exist.

```
!git clone https://github.com/username/repository.git
```

Then read file:

```
with open("/content/repository/data.txt", "r") as file:
    data = file.read()

print(data)
```

Example

Suppose file is:

https://github.com/user/demo/blob/main/names.txt

Raw URL:

https://raw.githubusercontent.com/user/demo/main/names.txt

Read in Colab:

```
import requests

url = "https://raw.githubusercontent.com/user/demo/main/names.txt"

data = requests.get(url).text

print(data)
```

Output Example

```
Rahul
Amit
Neha
Riya
```